MATH.APP.220 Multivariable Calculus

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Vector-valued functions

- A vector-valued function $F : D \to \mathbb{R}^m$, where $m > 1$ and $D \subset \mathbb{R}^n$, maps an element $\mathbf{x} \in \mathbb{R}^n$ to an element $F(\mathbf{x}) \in \mathbb{R}^m$.
- Real-valued functions f_i, $i = 1, 2, \ldots, m$ are the **component functions** of F, i.e.

$$F(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_m(\mathbf{x})).$$

- For example, a **polar coordinate mapping** $F : \mathbb{R}^2 \to \mathbb{R}^2$,

$$F(r, \theta) = (r\cos\theta, r\sin\theta)$$

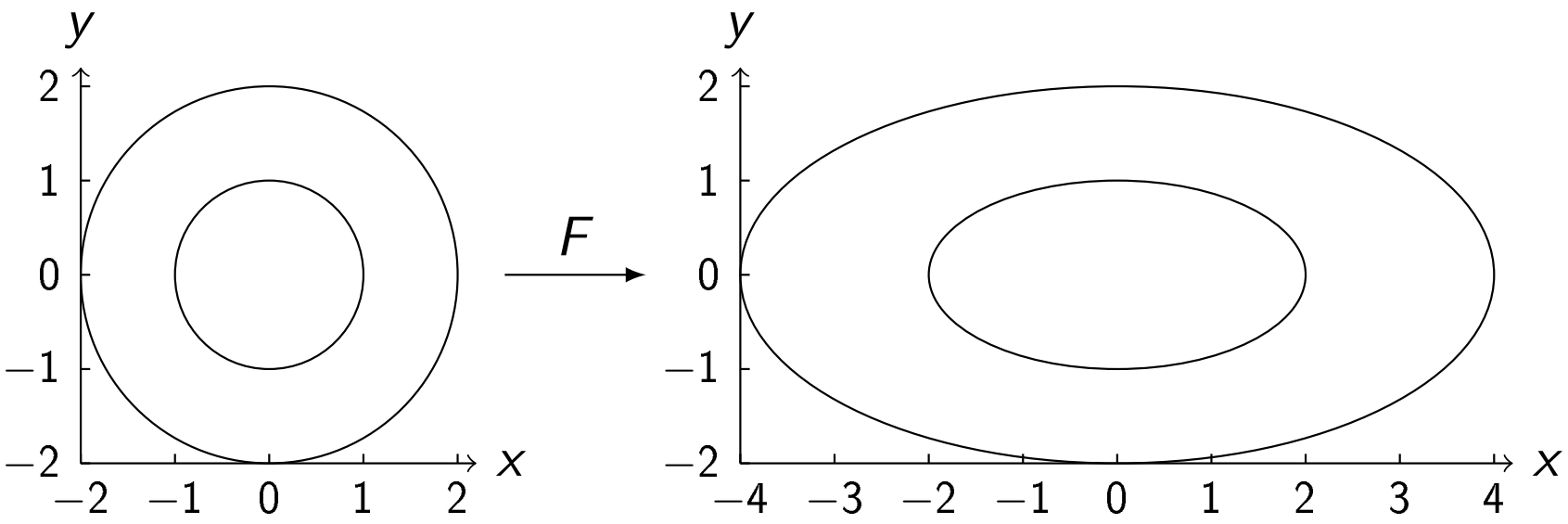
will map a point given in polar coordinates to a point in cartesian coordinates $x = r\cos\theta$ and $y = r\sin\theta$.

Visualization as a pair of pictures

- We have the possibility to try to visualize a vector-valued function, for which $n \leq 3$ and $m \leq 3$.
- There's an example of our first way of visualization next page. A function of the type $\mathbb{R}^2 \to \mathbb{R}^2$ can be visualized by plotting some curves C_i contained in the domain into a picture of the xy-plane, and next to it another picture containing the images $F(C_i)$ of the curves in the range, also subset of the xy-plane.
- The same technique could be used (on a computer, mostly) to visualize functions of the type $\mathbb{R}^3 \to \mathbb{R}^3$, with surfaces contained in the domain and their images in the range of F.

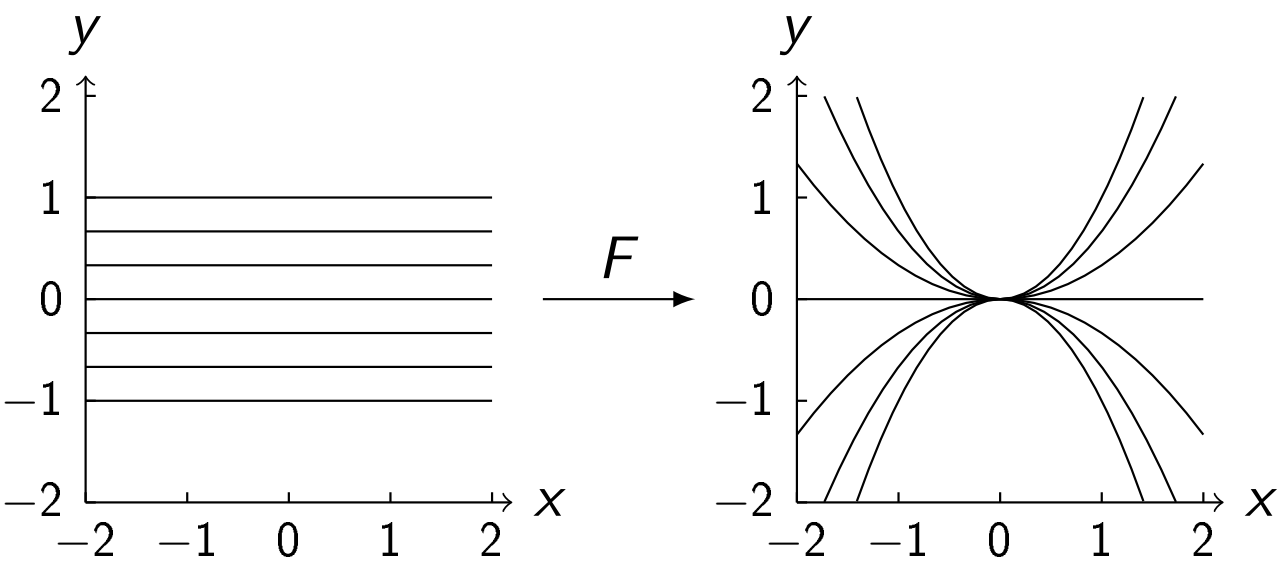
Visualization as a pair of pictures

The function $F(x,y) = (2x, y)$ is visualized below using circles plotted in the domain. When the x-coordinates of the points on a circle are doubled, we get ellipses.



Visualization as a pair of pictures

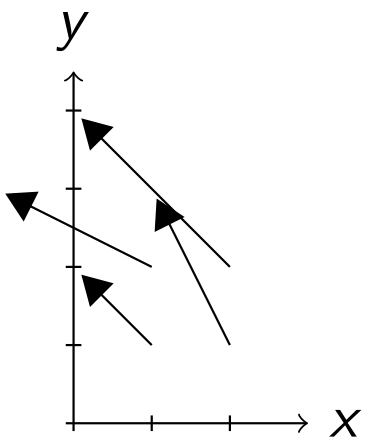
The function $F(x,y) = (x, x^2 y)$ is visualized below using horizontal lines. The line $y = c$ maps to the curve $y = cx^2$. The mapping is pushing the points of the lines towards the x-axis when $-1 < x < 1$, and away from the x-axis elsewhere.



Vector field

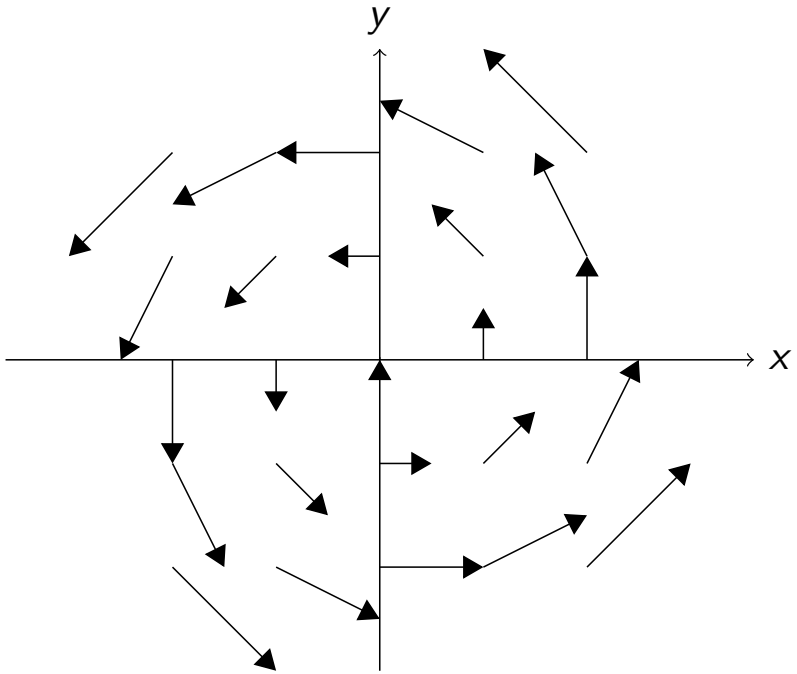
- In the cases $F : \mathbb{R}^2 \to \mathbb{R}^2$ and $F : \mathbb{R}^3 \to \mathbb{R}^3$, the function F can be interpreted as a **vector field** that can be visualized by drawing the vectors $F(\mathbf{x})$ in the points $\mathbf{x}$ that are at a suitable distance from each other.
- Evaluate the function F ar some chosen points and plot the associated vectors starting from those points.
- For example, $F(x,y) = (-y,x)$.

(x,y)	$F(x,y)$
$(1,1)$	$(-1,1)$
$(1,2)$	$(-2,1)$
$(2,1)$	$(-1,2)$
$(2,2)$	$(-2,2)$



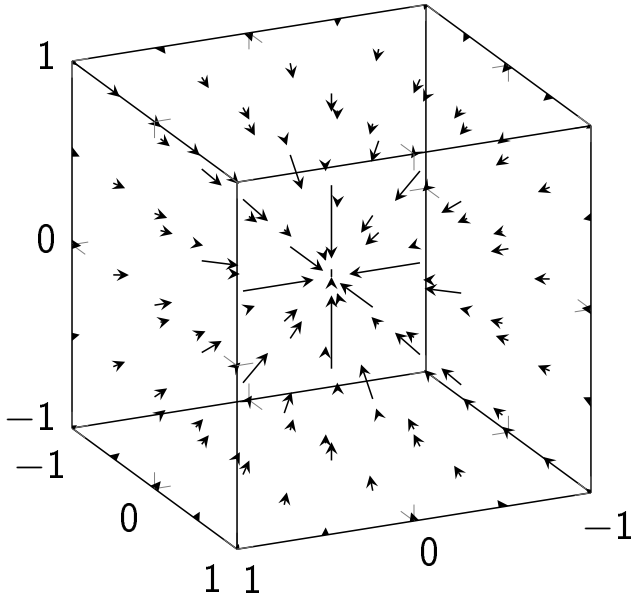
Vector field

Plotting just a couple of vectors does not give a clear picture of how the field behaves. The pictures are often plotted on a computer. Below is a another picture of $F(x,y) = (-y,x)$ with more coverage. The lengths of the vectors are scaled by the factor 0.5, so that the vectors would not go on top of each other.



The electric field of a negative point charge

Depicted is the field $E(\mathbf{x}) = -c\frac{\mathbf{x}}{\|\mathbf{x}\|^3} = -\frac{c}{\|\mathbf{x}\|^2}\frac{\mathbf{x}}{\|\mathbf{x}\|}$. A vector of the form $\frac{\mathbf{x}}{\|\mathbf{x}\|}$ has length 1 and is directed away from the origin. With a minus sign, it is directed towards the origin. The magnitude of the electric field is $\frac{c}{\|\mathbf{x}\|^2}$. This tells us that the magnitude of the field is inversely proportional to the square of the distance.



Definition of a limit

- Let $D \subset \mathbb{R}^n$. The function $f : D \to \mathbb{R}^m$ has a limit $L \in \mathbb{R}^m$ at an accumulation point $\mathbf{a} \in \overline{D}$, if for each $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$\mathbf{x} \in D \text{ and } 0 < \|\mathbf{x} - \mathbf{a}\| < \delta \quad \Rightarrow \quad \|F(\mathbf{x}) - L\| < \varepsilon.$$

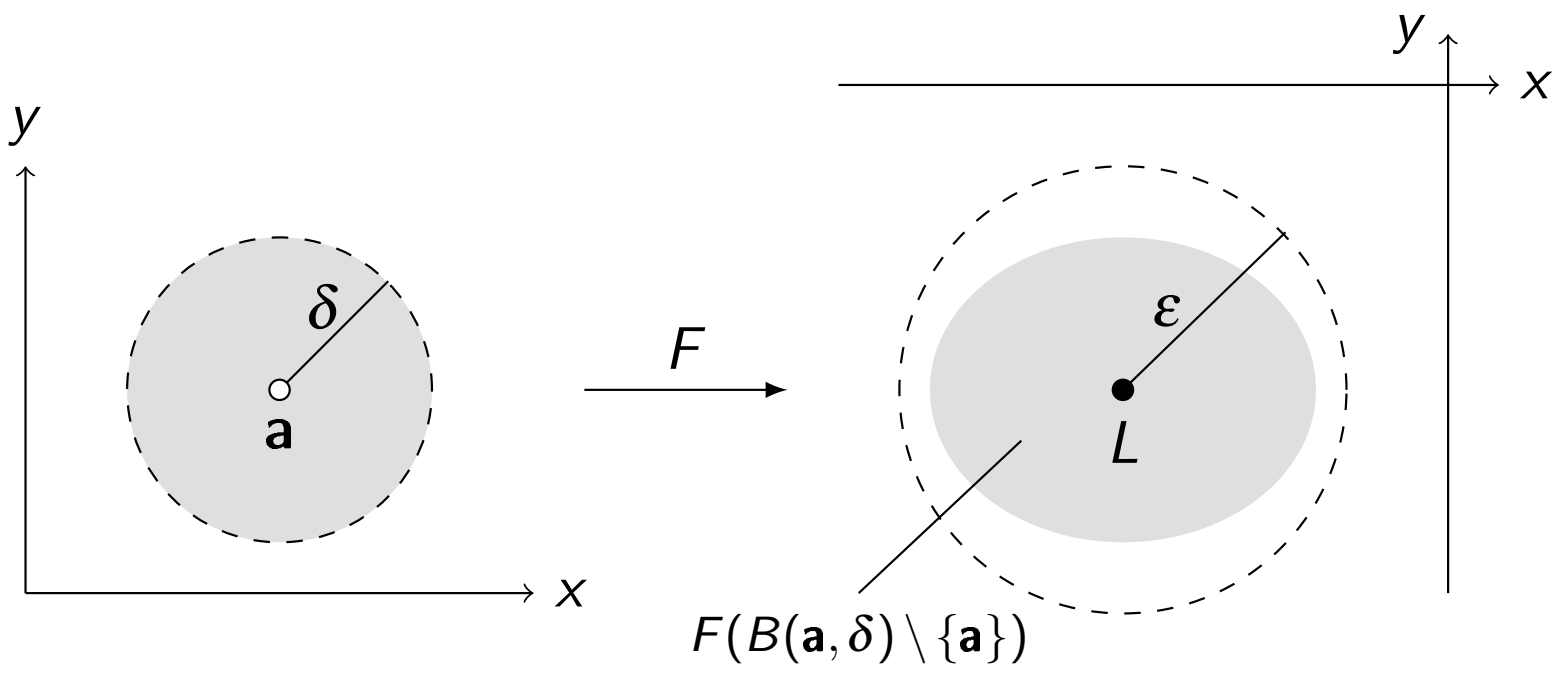
- If the limit exists, denote $\lim_{\mathbf{x} \to \mathbf{a}} F(\mathbf{x}) = L$ or $F(\mathbf{x}) \to L$, when $\mathbf{x} \to \mathbf{a}$.
- The limit is evaluated by evaluating the limit of each component function,

$$\lim_{\mathbf{x} \to \mathbf{a}} F(\mathbf{x}) = \left(\lim_{\mathbf{x} \to \mathbf{a}} f_1(\mathbf{x}), \lim_{\mathbf{x} \to \mathbf{a}} f_2(\mathbf{x}), \ldots, \lim_{\mathbf{x} \to \mathbf{a}} f_m(\mathbf{x}) \right).$$

Visualization of the condition in the definition

The condition in the definition means that an open ball centered at $\mathbf{a}$ of radius δ is mapped to an open ball centered at L of radius ε, i.e.

$$F(B(\mathbf{a},\delta)\setminus\{\mathbf{a}\}) \subset B(L,\varepsilon).$$



Definition of continuity

- Let $D \subset \mathbb{R}^n$ and $\mathbf{a} \in D$ an accumulation point of D. The function $f : D \to \mathbb{R}$ is **continuous at a**, if

$$F(\mathbf{a}) = \lim_{\mathbf{x} \to \mathbf{a}} F(\mathbf{x}).$$

- If f is continuous at every point in D, then F is **continuous**.
- For non-accumulation points $\mathbf{a} \in D$, F is continuous at $\mathbf{a}$.
- A function F is continuous at $\mathbf{a}$ if and only if each component function f_i, $i = 1, 2, \ldots, m$, of F is continuous at $\mathbf{a}$.

Reasoning when justifying a function is continuous

- Let $D \subset \mathbb{R}^n$. If $\lambda \in \mathbb{R}$ and the functions $F : D \to \mathbb{R}^m$ and $G : D \to \mathbb{R}^m$ are continuous at $\mathbf{a}$, then also the functions $F + G$ and λF are continuous at $\mathbf{a}$.
- Let $D \subset \mathbb{R}^n$. If $G : D \to \mathbb{R}^p$ is continuous at $\mathbf{a} \in D$ and $F : G(D) \to \mathbb{R}^m$ is continuous at $G(\mathbf{a}) \in \mathbb{R}^p$, then the composite function $F \circ G : D \to \mathbb{R}^m$, $(F \circ G)(\mathbf{x}) = F(G(\mathbf{x}))$ is continuous at $\mathbf{a}$.

$$\underset{\mathbf{a}}{\overset{\mathbb{R}^n}{\bullet}} \xrightarrow{\quad G \quad} \underset{G(\mathbf{a})}{\overset{\mathbb{R}^p}{\bullet}} \xrightarrow{\quad F \quad} \underset{F(G(\mathbf{a}))}{\overset{\mathbb{R}^m}{\bullet}}$$

Derivative matrix

- Consider $F = (f_1, f_2, \ldots, f_n) : D \to \mathbb{R}^m$, where $D \subset \mathbb{R}^n$. Assume the component functions f_i have all of their partial derivatives $\frac{\partial f_j}{\partial x_i}$ at $\mathbf{x}$. The **derivative matrix** of F (the **Jacobian matrix**) at $\mathbf{x}$ is the $m \times n$-matrix (also denoted $DF(\mathbf{x})$)

$$F'(\mathbf{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1}(\mathbf{x}) & \frac{\partial f_1}{\partial x_2}(\mathbf{x}) & \cdots & \frac{\partial f_1}{\partial x_n}(\mathbf{x}) \\ \frac{\partial f_2}{\partial x_1}(\mathbf{x}) & \frac{\partial f_2}{\partial x_2}(\mathbf{x}) & \cdots & \frac{\partial f_2}{\partial x_n}(\mathbf{x}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(\mathbf{x}) & \frac{\partial f_m}{\partial x_2}(\mathbf{x}) & \cdots & \frac{\partial f_m}{\partial x_n}(\mathbf{x}) \end{bmatrix}.$$

Jacobian determinant

- The determinant of the derivative matrix of $F : \mathbb{R}^n \to \mathbb{R}^n$ is called the **Jacobian determinant** $J_F(\mathbf{x}) = \det(F'(\mathbf{x}))$.
- Let's calculate the Jacobian determinant of the polar coordinate mapping $F(r, \theta) = (r\cos\theta, r\sin\theta)$ as an example. The derivative matrix is

$$F'(r, \theta) = \begin{bmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{bmatrix}.$$

- Therefore, the Jacobian determinant is

$$\begin{aligned} J_F(r, \theta) &= \cos\theta \cdot r\cos\theta - (-r\sin\theta) \cdot \sin\theta \\ &= r\left(\cos^2\theta + \sin^2\theta\right) = r. \end{aligned}$$

Significance of the Jacobian determinant

- We can approximate the function $f : \mathbb{R} \to \mathbb{R}$ near the point a by the tangent line with slope $f'(a)$. For a small interval I containing a, the length of $f(I)$ is approximately $|f'(a)|$ times the length of I.
- The function $F : \mathbb{R}^2 \to \mathbb{R}^2$ can be approximated near the point $\mathbf{a}$ by the linear mapping $T(\mathbf{x}) = F(\mathbf{a}) + F'(\mathbf{a})(\mathbf{x} - \mathbf{a})$.
- In this mapping, the image of the set $A \subset \mathbb{R}^2$, where A contains $\mathbf{a}$, is $T(A)$. The area of the set $T(A)$ is approximately $|\det F'(\mathbf{a})|$ times the area of A.
- Therefore, the area of the image $F(A)$ of the set A containing $\mathbf{a}$ is approximately $|J_F(\mathbf{a})|$ times the area of A.

Differentiability of vector-valued functions

- The function $F : D \to \mathbb{R}^m$, where $D \subset \mathbb{R}^n$, is **differentiable at** $\mathbf{a} \in \mathbb{R}^n$, if all component functions of F have all their partial derivatives at $\mathbf{a}$ and if

$$F(\mathbf{a}+\mathbf{h}) = F(\mathbf{a}) + F'(\mathbf{a})\mathbf{h} + \varepsilon(\mathbf{h})\|\mathbf{h}\|,$$

  where $\varepsilon(\mathbf{h}) \to 0$, when $\mathbf{h} \to \mathbf{0}$. Furthermore, F is **differentiable**, if it is differentiable at every point of D.

- The **linearization** of F at $\mathbf{a}$ is the function $T : D \to \mathbb{R}$,

$$T(\mathbf{x}) = F(\mathbf{a}) + F'(\mathbf{a})(\mathbf{x}-\mathbf{a})$$

  and $F(\mathbf{x}) \approx T(\mathbf{x})$ is the **linear approximation** of F at $\mathbf{a}$.

Differentiability of vector-valued functions

- Differentiability and linear approximation could not exists without the derivative matrix. Namely we have the theorem:
- The function $F : D \to \mathbb{R}^m$, where $D \subset \mathbb{R}^n$, is differentiable at $\mathbf{a}$, if and only if there exists an $m \times n$-matrix M such that

$$F(\mathbf{a}+\mathbf{h}) = F(\mathbf{a}) + M\mathbf{h} + \varepsilon(\mathbf{h})\|\mathbf{h}\|,$$

  where $\varepsilon(\mathbf{h}) \to 0$, when $\mathbf{h} \to \mathbf{0}$. Then $M = F'(\mathbf{a})$.
- Differentiability could be proven by checking that

$$\lim_{\mathbf{h}\to\mathbf{0}} \frac{F(\mathbf{a}+\mathbf{h}) - F(\mathbf{a}) - F'(\mathbf{a})\mathbf{h}}{\|\mathbf{h}\|} = \mathbf{0}.$$

Theorems involving differentiability

- A differentiable function $F : D \to \mathbb{R}^m$, where $D \subset \mathbb{R}^n$, is continuous.
- The function $F = (f_1, f_2, \ldots, f_m) : D \to \mathbb{R}^m$, where $D \subset \mathbb{R}^n$, is differentiable at $\mathbf{a} \in \mathbb{R}^n$ if and only if each component function f_j is differentiable at $\mathbf{a}$.
- If all partial derivatives $\frac{\partial f_j}{\partial x_i}$ of all component functions f_j of $F : D \to \mathbb{R}^m$, $D \subset \mathbb{R}^n$, are continuous, then F is **continuously differentiable**.
- If F is continuously differentiable, then F is differentiable.

Chain rule

- Previous course: If g is differentiable at a and f is differentiable at $g(a)$, then the composite function $h(x) = (f \circ g)(x) = f(g(x))$ is differentiable at a and the derivative is
$$h'(a) = f'(g(a))\, g'(a).$$
- Now: Let $G : D \to \mathbb{R}^p$, where $D \subset \mathbb{R}^n$, be differentiable at $\mathbf{a} \in \mathbb{R}^n$ and $F : G(D) \to \mathbb{R}^m$ be differentiable at $G(\mathbf{a}) \in \mathbb{R}^p$. Then the composite function $H = F \circ G : D \to \mathbb{R}^m$, $D \subset \mathbb{R}^n$, $H(\mathbf{x}) = F(G(\mathbf{x}))$ is differentiable at $\mathbf{a}$ and
$$H'(\mathbf{a}) = F'(G(\mathbf{a}))\, G'(\mathbf{a}).$$

Example cases of chain rule: $n = m = 1$ and $p = 3$

- Let $\mathbf{r} : \mathbb{R} \to \mathbb{R}^3$, $\mathbf{r}(t) = (x(t), y(t), z(t))$ and $f : \mathbb{R}^3 \to \mathbb{R}$, $f = f(x, y, z)$. Then the composite function $w(t) = f(\mathbf{r}(t))$ is a real-valued single variable function and the chain rule appears as

$$\frac{dw}{dt} = f'(\mathbf{r}(t))\mathbf{r}'(t) = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{bmatrix} \begin{bmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \\ \frac{dz}{dt} \end{bmatrix},$$

i.e.

$$\frac{dw}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} + \frac{\partial f}{\partial z}\frac{dz}{dt}.$$

Example cases of chain rule: $n = 3$ and $p = m = 1$

- Let $u : \mathbb{R}^3 \to \mathbb{R}$, $u = u(x, y, z)$ and $f : \mathbb{R} \to \mathbb{R}$, $f = f(u)$. The derivative matrix of $w(x, y, z) = f(u(x, y, z))$ is

$$\begin{aligned} w'(x,y,z) &= f'(u(x,y,z))\, u'(x,y,z) \\ &= \begin{bmatrix} \dfrac{df}{du} \end{bmatrix} \begin{bmatrix} \dfrac{\partial u}{\partial x} & \dfrac{\partial u}{\partial y} & \dfrac{\partial u}{\partial z} \end{bmatrix} \\ &= \begin{bmatrix} \dfrac{df}{du}\dfrac{\partial u}{\partial x} & \dfrac{df}{du}\dfrac{\partial u}{\partial y} & \dfrac{df}{du}\dfrac{\partial u}{\partial z} \end{bmatrix}, \end{aligned}$$

in other words, the partial derivatives are

$$\frac{\partial w}{\partial x} = \frac{df}{du}\frac{\partial u}{\partial x}, \qquad \frac{\partial w}{\partial y} = \frac{df}{du}\frac{\partial u}{\partial y}, \quad \text{ja} \quad \frac{\partial w}{\partial z} = \frac{df}{du}\frac{\partial u}{\partial z}.$$

Example cases of chain rule: $n = 2$, $p = 3$ and $m = 1$

- Let $G : \mathbb{R}^2 \to \mathbb{R}^3$, $G(u,v) = (x(u,v), y(u,v), z(u,v))$ and $f : \mathbb{R}^3 \to \mathbb{R}$, $f = f(x,y,z)$. The derivative matrix of $w(u,v) = f(G(u,v))$ is

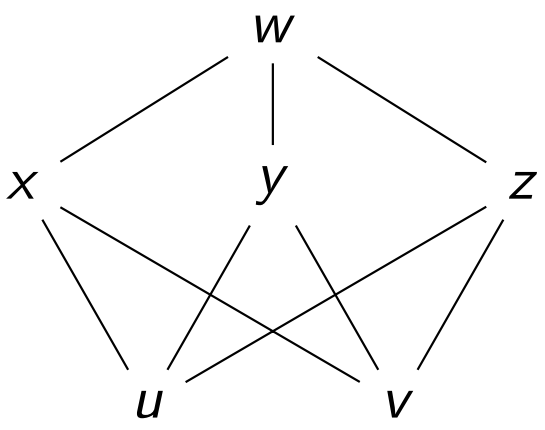
$$
\begin{aligned}
w'(u,v) &= f'(G(u,v))\, G'(u,v) \\
&= \begin{bmatrix} \dfrac{\partial f}{\partial x} & \dfrac{\partial f}{\partial y} & \dfrac{\partial f}{\partial z} \end{bmatrix} \begin{bmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \\ \dfrac{\partial z}{\partial u} & \dfrac{\partial z}{\partial v} \end{bmatrix} \\
&= \begin{bmatrix} \dfrac{\partial f}{\partial x}\dfrac{\partial x}{\partial u} + \dfrac{\partial f}{\partial y}\dfrac{\partial y}{\partial u} + \dfrac{\partial f}{\partial z}\dfrac{\partial z}{\partial u} & \dfrac{\partial f}{\partial x}\dfrac{\partial x}{\partial v} + \dfrac{\partial f}{\partial y}\dfrac{\partial y}{\partial v} + \dfrac{\partial f}{\partial z}\dfrac{\partial z}{\partial v} \end{bmatrix}.
\end{aligned}
$$

Example cases of chain rule: $n = 2$, $p = 3$ and $m = 1$

- The partial derivatives of the case presented on the previous page are

$$\frac{\partial w}{\partial u} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial u} + \frac{\partial f}{\partial z}\frac{\partial z}{\partial u}$$
$$\frac{\partial w}{\partial v} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial v} + \frac{\partial f}{\partial z}\frac{\partial z}{\partial v}.$$

- The following tree diagram depicts this chain rule:



Directional derivative

- Earlier on this course it was mentioned that the partial derivatives are interpreted as derivatives to the function in the directions of the coordinate axes.
- We can also take the derivative in other directions than the coordinate axes. In this case we call them **directional derivatives**.
- The directional derivative of $f : D \to \mathbb{R}$, where $D \subset \mathbb{R}^n$, at $\mathbf{x} \in \mathbb{R}^n$ along (=in the direction of) the **unit** vector $\mathbf{e} \in \mathbb{R}^n$ is

$$D_{\mathbf{e}} f(\mathbf{x}) = \lim_{h \to 0} \frac{f(\mathbf{x} + h\mathbf{e}) - f(\mathbf{x})}{h}.$$

Calculating a directional derivative

- If the function $f : D \to \mathbb{R}$, where $D \subset \mathbb{R}^n$, is differentiable at $\mathbf{x} \in \mathbb{R}^n$, then the directional derivative $D_{\mathbf{e}} f(\mathbf{x})$ exists in every direction $\mathbf{e} \in \mathbb{R}^n$, $\|\mathbf{e}\| = 1$, and is calculated as

$$D_{\mathbf{e}} f(\mathbf{x}) = f'(\mathbf{x})\,\mathbf{e}.$$

- When calculating the directional deriative, it is essential that $\mathbf{e}$ is a unit vector. If necessary, normalize the given direction (=divide it by its norm).
- The opposite does not hold. A function may have a directional derivative at $\mathbf{x}$ in every direction, but f might still not be differentiable at $\mathbf{x}$.

An example of a directional derivative

- Calculate the directional derivative of $f(x,y) = 3x^2y$ at $(1,2)$ along the vector $(3,4)$.
- The norm of the direction vector is $\sqrt{3^2+4^2} = \sqrt{25} = 5$, and thus the unit direction is $\mathbf{e} = \left(\frac{3}{5}, \frac{4}{5}\right)$.
- The directional derivative along the vector $(3,4)$ at (x,y) is

$$D_{(3,4)}f(x,y) = \begin{bmatrix} 6xy & 3x^2 \end{bmatrix} \begin{bmatrix} \frac{3}{5} \\ \frac{4}{5} \end{bmatrix} = \frac{18xy + 12x^2}{5}.$$

- At $(1,2)$ we get

$$D_{(3,4)}f(1,2) = \frac{1}{5}\left(18 \cdot 1^2 \cdot 2 + 12 \cdot 1^2\right) = \frac{48}{5}.$$

Gradient

- Let the function $f : D \to \mathbb{R}$, where $D \subset \mathbb{R}^n$, have all partial derivatives at $\mathbf{x}$. The **gradient** of f at $\mathbf{x}$ is the vector

$$\nabla f(\mathbf{x}) = \left(\frac{\partial f}{\partial x_1}(\mathbf{x}), \frac{\partial f}{\partial x_2}(\mathbf{x}), \ldots, \frac{\partial f}{\partial x_n}(\mathbf{x}) \right).$$

- Note that the gradient is a column vector that contains the same elements as the derivative matrix, which is a row vector for a real-valued function. ∇ is said "nabla".
- Directional derivative and chain rule ($n = m = 1$, $p = 3$):

$$D_{\mathbf{e}} f(\mathbf{x}) = \nabla f(\mathbf{x}) \cdot \mathbf{e},$$
$$\frac{d}{dt} f(\mathbf{r}(t)) = \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t)$$

Geometric interpretation of the gradient

- Let $f : D \to \mathbb{R}$, where $D \subset \mathbb{R}^n$, be differentiable at $\mathbf{x}$. Then the maximum value of the directional derivative of f at $\mathbf{x}$ is $\|\nabla f(\mathbf{x})\|$.
- The function f increases the fastest in the direction of the gradient $\nabla f(\mathbf{x})$, i.e. the unit direction $\mathbf{e} = \dfrac{\nabla f(\mathbf{x})}{\|\nabla f(\mathbf{x})\|}$, and the maximum increase is $\|\nabla f(\mathbf{x})\|$.
- The function f decreases the fastest in the direction opposite to the gradient, $-\nabla f(\mathbf{x})$, and the maximum decrease is $-\|\nabla f(\mathbf{x})\|$.
- In a direction that is perpendicular to the gradient, the local rate of change of f is 0.

The gradient's relation to contour sets

- If $f : D \to \mathbb{R}$, where $D \subset \mathbb{R}^2$, is a continuously differentiable function and $\mathbf{r}(t)$ the parametrization of a smooth curve of the contour set (in $\mathbb{R}^2$ this means a level curve)

$$f^{-1}(c) = \left\{(x, y) \in \mathbb{R}^2 : f(x, y) = c\right\},$$

then for each $t \in D$ it holds that

$$\nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = 0.$$

- Geometrically this means that the gradient is always perpendicular to the level curve, a.k.a. contour curve.
- Similarly for $f : D \to \mathbb{R}$, $D \subset \mathbb{R}^3$, the gradient is always perpendicular to a contour surface.

Finding tangent lines based on contour interpretation

- If $\mathbf{n}$ is a normal of a line, $\mathbf{x}$ an arbitrary point and $\mathbf{p}$ a known point on the line, then the line given in the normal form is

$$(\mathbf{x}-\mathbf{p})\cdot\mathbf{n}=0.$$

- For example, the ellipse $x^2-xy+y^2=7$ can be interpreted as a level curve of $f(x,y)=x^2-xy+y^2$ on which $f(x,y)=7$. The gradient $\nabla f=(2x-y,2y-x)$ can be interpreted to be a normal of the ellipse (and its tangent) at the point (x,y).
- The equation of the tangent line to the ellipse at $(-1,2)$ is

$$((x,y)-(-1,2))\cdot(2(-1)-2,2\cdot 2-(-1))=0,$$

i.e. $4x-5y=-14$.

Constrained extrema

- Finding a **constrained extremum** means finding an extremum of f in a set smaller than the domain. The set is bounded by one or more **constraints**, normally an equation or equations that define the smaller set.
- Consider extrema in a triangle. We first look for the critical points. For that we don't have constraints, it's the whole domain we search from. On one of the three boundary lines, the constraint on that line is the equation of the line.
- We now have a choice. We can look for constrained extrema as before, plugging the equation of a boundary curve in the function etc., or alternatively we can use **Lagrange's method**.

Lagrange's method for $f(x,y)$, one constraint

- When searching for the local extrema of $f(x,y)$ in a set given by the constraint $g(x,y)=0$, where the functions f and $g: D \to \mathbb{R}$, $D \subset \mathbb{R}^2$, are continuously differentiable. Local extrema are either the solutions of the system

$$\begin{cases} g(x,y)=0 \\ D_x f(x,y) = \lambda D_x g(x,y) \\ D_y f(x,y) = \lambda D_y g(x,y) \end{cases}$$

  OR the points (x,y), for which $g(x,y)=0$ and $\nabla g(x,y)=(0,0)$.

- The variable λ is called the **Lagrange's multiplier**. No need to solve it from the system, we just want the points (x,y).

The Hessian

- The **Hessian** of the function $f : D \to \mathbb{R}$, where $D \subset \mathbb{R}^n$, at $\mathbf{x}$ is

$$H_f(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2}(\mathbf{x}) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(\mathbf{x}) \\ \frac{\partial^2 f}{\partial x_2 \partial x_1}(\mathbf{x}) & \frac{\partial^2 f}{\partial x_2^2}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n}(\mathbf{x}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(\mathbf{x}) & \frac{\partial^2 f}{\partial x_n \partial x_2}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_n^2}(\mathbf{x}) \end{bmatrix}.$$

- The Hessian can be used, for example, to prove the second derivative test for classifying local extrema that we had earlier.

Classifying local extrema

- Let the function $f : D \to \mathbb{R}$, where $D \subset \mathbb{R}^n$, have all second order partial derivatives at the critical point $\mathbf{a}$. Consider the Hessian $H_f(\mathbf{a})$.
  - If all eigenvalues of $H_f(\mathbf{a})$ are positive, $\mathbf{a}$ is a local minimum.
  - If all eigenvalues of $H_f(\mathbf{a})$ are negative, $\mathbf{a}$ is a local maximum.
  - If some eigenvalues are positive and some negative, $\mathbf{a}$ is a saddle point.
  - If zero is an eigenvalue, the test is inconclusive. Try classifying $\mathbf{a}$ in some other way.
- On this course, we mostly use Matlab to find the eigenvalues.